The Laureate text-to-speech system — architecture and applications

J H Page and A P Breen

The synthesis of high-quality, natural-sounding speech from text is an increasingly important requirement in many telecommunications applications. It enables customers to easily access text-based information, anywhere, any time, using only an ordinary fixed or mobile telephone of which there are more than 26 million in the UK and over half a billion world-wide. This paper describes several applications of text-to-speech synthesis and reviews the factors which should be considered when choosing text-to-speech synthesis for an application. Laureate, BT's text-to-speech system, has a structure designed to support rapid and cost-effective downstreaming to products, while remaining flexible enough to accommodate the latest techniques in synthesis. This unique architecture is described, along with the ways it can be tailored to specific applications. The results of speech quality measurements of the Laureate system are presented.

1. Introduction

Making machines that can talk has been a goal of engineers and scientists for many decades. The speaking clock, which was produced by the Research Branch of the British Post Office in 1936 [1], was one of the first systems to address this dream, and used analogue recording and playback techniques. The speech was stored using an optical method on twelve inch glass discs (Fig 1).

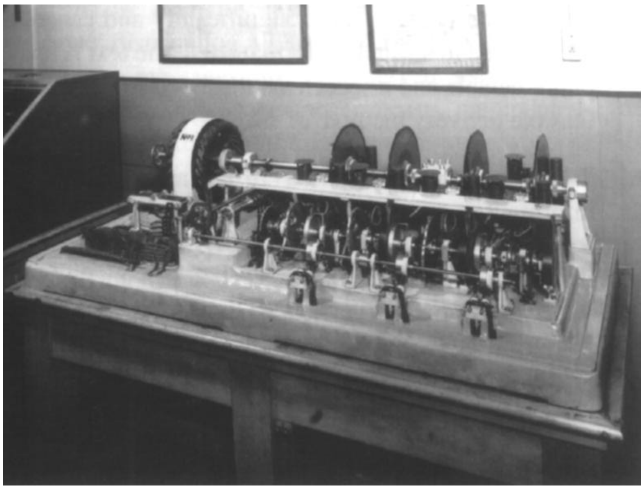

Fig 1 Picture of 1936 speaking clock.

This system was very sophisticated for the 1930s because, as well as getting the time right, it used concatenated speech to avoid recording all the material for the complete 24-hour cycle. With the technology then available, that would have taken nearly 8 miles of sound track! With improvements in recording techniques, network announcements and recorded information lines have become commonplace. However, it was not until the development of digital recording that it became possible to easily manipulate recorded speech on computer systems. Prior to that, the use of continuously running looped tape systems, for recorded announcements, resulted in callers being connected at any point within the message. Today, sophisticated interactive systems, capable of extensive dialogues using fixed recorded messages, are being deployed. However, when there is a need to utter a variable piece of speech, it is very difficult to make it sound natural with simple recordings and concatenation techniques. In such circumstances, the synthesis of natural sounding speech from a textual input becomes a desirable capability, but one that is several orders of magnitude harder to achieve than speaking a prerecorded message.

Several factors complicate the process, but the main one is that the media of speech and text have quite different purposes. Speech is the primary **human communication mechanism**, whereas the main purpose of text is to **record information**. The conversion from one medium to the other is rarely a one-to-one mapping — hence the work required to turn a book into a play.

Early attempts at synthesis were based on mechanical analogues of the human vocal tract (see Westall [2]), to be later replaced by sophisticated electronic models of the acoustic signal [3]. Provided that accurate parametric information on the speech signal was available, such models could produce natural-sounding synthetic speech. Unfortunately, these advances in the synthesis of speech were not matched by a corresponding increase in the understanding of how to automatically generate the appropriate parametric information to drive these synthesis models. As a consequence, the quality of synthetic speech generated from text sounded robotic and unnatural. With advances in computer technology, doubling processing power every 18 months or so, the possibilities for modelling, encoding and storing fragments of the speech signal have led to improvements in quality and a resurgence of interest in text-to-speech generation. However, the depiction of HAL, the talking computer in '2001' by Arthur C Clarke, still provides a quality target for text-to-speech synthesis systems.

Today, the quality of such synthetic speech has improved to such an extent that real world applications are now viable and the subject has moved from being a predominantly academic discipline to one of increasing importance to the telecommunications and computer industries. This shift in emphasis has led to a need by industry to evolve methods of research and development which enable novel ideas to be investigated, tested and developed, while simultaneously ensuring that products can be produced rapidly and cost effectively. The solution to this problem, adopted by BT Laboratories, is described later in this paper. But first, a description is given of some applications where the technology can produce some real advantages over other forms of speech output.

2. Applications for speech synthesis

The applications for a synthetic speech system which provides human quality speech are unlimited. Some of these applications place the synthesis at the point of use, e.g. talking machines for people who have lost their voices, or reading machines for the visually impaired. Many personal computers also have speech synthesis, and this use probably constitutes the single largest market for text to speech today. However, with the convergence of communications and computing technology, the use of text to speech within a communications system produces the greatest benefit. Using an ordinary telephone, of which there are more than 26 million in the UK and over half a billion worldwide, speech synthesis enables everyone to access network-based information sources.

The quality of the voice synthesiser influences the choice of application. Clearly, if the synthesiser output was equal to that of a good-quality recording, then any system requiring voice output would use it, if only for the generation of stored voice prompts. However, since the voice quality produced by text-to-speech synthesisers is, in general, not yet up to the standard of a good-quality recording, applications have to be carefully matched to the quality of the voice. To a certain extent, the choice of application depends on how important it is to the user to get the information, as well as its type and presentation. To illustrate this point, consider talking books for the blind and visually impaired. Most sighted people would choose to read a book themselves given a choice of delivery, but they may also choose to have a book read to them by another person. Blind and visually impaired people are also likely to prefer a human reader, but would probably make use of a speech synthesiser if they needed access to the information, and no recordings were available. However, some people prefer a synthesis system where they can control the speed of the speech, as this enables them to speed it up and absorb the information at a higher rate.

Another factor involved in matching text to speech (TTS) to the application is the quality of textual information. If the text consists of short sentences and phrases with good punctuation, then the quality of the synthesis will be perceived as being high. On the other hand, if the text is poorly conceived and difficult to read out loud for a human reader, then the synthesis quality will be similarly impaired.

2.1 When to use synthetic speech

There are a number of situations where synthetic speech is likely to be selected in preference to the simpler technology solutions for providing speech output.

- Where the text is unpredictable and dynamic

  Text-to-speech can be used in situations where the messages are short, typically a few lines, but the information content varies significantly and cannot be constrained to a standard format. Here the text is totally unpredictable and a speech synthesis system is the only feasible delivery method.

- Where access is required to a large database of information

  For large databases it is not feasible to store the information as recorded speech due to the costs of recording and storage. Large databases are rarely completely stable and this also favours a text-to-speech approach.

- Where the output is relatively stable, but cost of provision and lead time are critical

  An example of this is a telephone network announcement system where most of the announcements remain static, but situations can arise where new messages are necessary, often at short notice, and must be in the same voice. Voice synthesis greatly simplifies the process of generating fixed voice prompts for new services, as compared with live recordings which take a significant time to prepare.

- Where consistency of voice is a requirement

  Many services require the same voice for all announcements. This is not a problem if the service, once built, requires no future modifications. If, on the other hand, future enhancements are envisaged, there may be a problem with the availability of the original speaker. In this situation it may be better to choose a synthesised voice in the first place.

- Where low bandwidth is required

  By transmitting information in text form and converting to speech at the receiving end, only a very low bandwidth is used.

The next section gives more detailed descriptions of a number of applications which are well suited to using text-to-speech synthesis.

2.2 Example applications

Information lines

The importance and conciseness of the information is key to this type of service. With a suitably interactive dialogue (user input via speech recognition or TouchTone ® — dual tone multifrequency (DTMF) tones generated from the telephone keypad), the user can pinpoint the information required. This also means that the information can be provided in convenient chunks appropriate to the synthesis process.

There are currently many services available, over the telephone, distributing all kinds of information. Those with unchanging messages are best served using recordings. Other services, such as weather forecasts, are updated 2 to 3 times a day, and an automatic text-to-speech conversion system is a practical lower-cost option in these cases.

A prototype system, developed at BT Laboratories (BTL), demonstrates access to detailed, localised weather information. This is achieved by providing a system with forecast areas linked to dialling code areas. Access to the weather forecast for the region of interest is simply obtained by keying in the relevant dialling code.

Other types of information, such as stocks and share prices, vary continuously but have a fixed format. Here speech concatenation techniques can be appropriate. An example of a service giving the current price of some item might be:

‘*The price of gold is <n> dollars/ounce*’

where the main phrase is a single recording, but has the variable *n*, which will be one of a set of recorded numbers embedded in it. These concatenation systems can work well provided that the information content is simple. With more complex examples employing a larger number of fields with greater variability, the resulting speech quality becomes disconcerting with disjointed intonation and impairment.

The use of interactive systems, employing either speech recognition or DTMF tone detection for user input, will enable sophisticated information services to be provided. Here it will be possible to extract information from large databases which are too extensive to be provided in a pre-recorded form.

Automated catalogue ordering system

The extra information given to the customer, by using text to speech in this application, is likely to lead to a far higher level of customer confidence and satisfaction. It is also a good example of an application where information flows in both directions, as well as being an example of accessing a large database. Automated catalogue ordering systems can be provided with relatively simple technology, using DTMF tone detection to input the number of the item to be ordered and speech output via digit concatenation to confirm the correct receipt of this number. When TTS technology is used, a far more user friendly system can be constructed. With the TTS system the customer would enter the item number by speech or DTMF tone detection as before, but in this case the system responds with a full spoken description of the item ordered. This not only confirms that the correct item number has been recognised, but also that the customer had looked up and entered the correct item number from the catalogue. The TTS-based system would also read out the name and address to whom the order was to be sent, so that, at the end of the transaction, the customer would be confident that the right item had been ordered and that it would be sent to the right address.

Remote e-mail reading

Speech access to e-mail from any fixed or mobile telephone can only be achieved by using text-to-speech synthesis. It is particularly useful for people whose jobs take them away from their offices, but who do much of their communication via e-mail.

A prototype telephone access to e-mail system has been developed at BTL, which provides both an experimental platform and a demonstration system. A user interacts with the current prototype via a combination of DTMF tone detection and speech recognition. An advantage of using DTMF tones is that they are reliably detected in the presence of outgoing speech. Consequently, comprehensive prompts which have been provided for novices can be interrupted to give faster access for the experienced user. The facilities provided by the system are as follows:

- PIN code security,
- review of messages by sender or subject,
- read-out of e-mail with the facility to go back, repeat or go forward,
- facility to delete messages,
- facility to send messages by facsimile to any destination.

Additional facilities which would be provided in a fully featured system are:

- voice mail reply with an e-mail notification,
- facility to file messages.

A highly desirable feature is the facility to provide a spoken reply to e-mails, with the speech converted to text using speech recognition. However, since current speaker-independent recognition systems can only have a restricted vocabulary [4], a compromise solution of a voice mail reply along with an e-mail notification of the voice mail, would be a useful future extension.

Personal navigation systems

With the advent of inexpensive GPS (Global Positioning System) receivers [5] and mobile phones, it is possible to provide navigational guidance facilities, with one basic arrangement, for a variety of navigation-related applications. As shown in Fig 2, this consists of a GPS receiver combined with a mobile phone linking back to a central computer located within the telephone network. The mobile phone transmits the current position from the GPS receiver to the central computer which then reads directions via a text-to-speech synthesiser to direct the user to the previously entered destination.

Other navigation systems link a GPS receiver directly to an on-board computer which stores map and routeing information; but the advantage of using a network-based computer for the information storage is that it can be constantly updated with any relevant information. This would be particularly useful in a car navigation system where varying road traffic information could be taken into account when giving directions. Speech-based systems also have obvious safety advantages over in-car electronic map systems which require the driver to take their eyes off the road.

Blind and visually impaired people could also benefit from a similar system, by using the more accurate differential GPS systems [6]. These are sufficiently accurate to guide them along the street avoiding all fixed objects such as lamp posts.

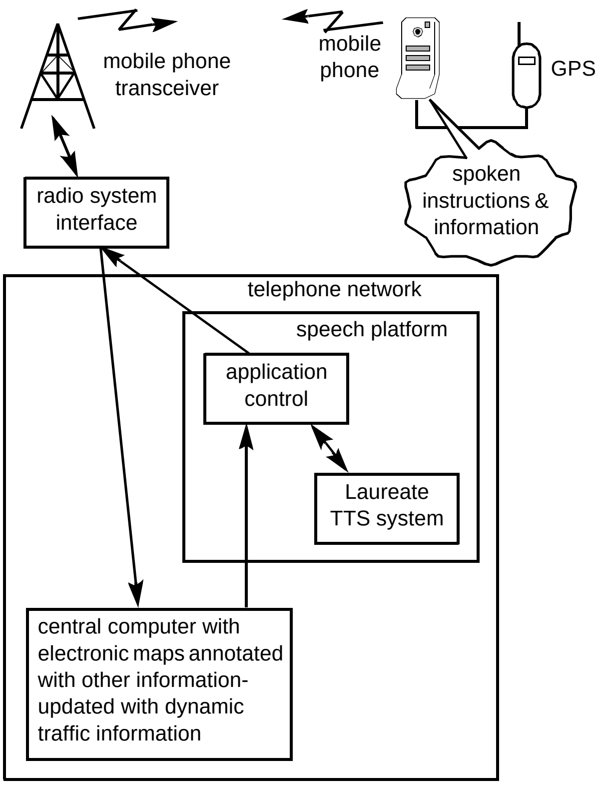

Fig 2 Personal navigation and information system.

Navigation systems of this type are universally applicable, providing people with voice-based personal guidance systems which would be just as useful navigating a route in London as it would when out walking in the countryside. It would even be possible to have guided countryside walks along with commentaries about interesting features along the route. Guides to the local pubs and restaurants could also be included.

Natural language or conversational interactive voice response (IVR) systems

One area of current IVR research (expanded on by Wyard [7]) is directed towards making systems more conversational, allowing the user to speak to the machine in a more natural manner, unconstrained by today's menu-driven approach. Natural-language systems will almost inevitably require the use of on-line text-to-speech synthesis to utilise the greater variety of response provided.

Synthetic persona

This application provides a lifelike synthetic talking face (Fig 3). The head and shoulder images are generated using a polygonal wire-frame model onto which the image of a person is mapped [8]. If the wire frame is rotated or translated and the stored image texture projected into a moving sequence of images, a convincing 3-D effect is produced. This is achieved despite only two images being used initially for the texture mapping.

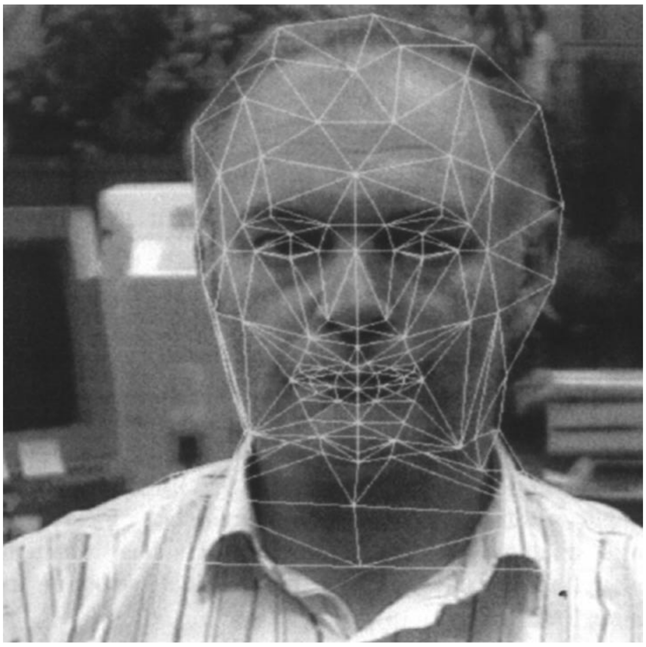

Fig 3 Picture of synthetic persona.

Moreover, by transforming the positions of sets of the vertices of the wire-frame model resulting in distortion of the surface texture, synthesised facial expressions can be generated. Linking this synthetic facial image of a talker to the synthetic speech of the same talker produces a very compelling and entertaining way of providing computer-based information. The basic source data driving the system is computer stored text, but appears to be delivered to the user by a human presenter.

This arrangement can be used to provide a more personal, human interface, to network services, an example being an automated announcer for video on demand services. By providing suitable helper applications to World WideWeb (WWW) browsers, this type of system could also be used for the presentation of information in an interesting way.

Laureate, BT's text-to-speech system, which has an architecture designed to support these communications applications, is described in the next section.

3. Architecture of the Laureate text-to-speech system

A simple description of a text-to-speech system and the technical challenges introduces this section. Subsequently, the philosophy and structure of the Laureate system architecture is presented. Two companion papers describe the algorithms employed in text-to-speech systems. The first deals with text and linguistic analysis [9] and the second, with prosody and speech generation [10].

3.1 Technical challenges

The process of text-to-speech conversion can be broken into three main parts, as shown in Fig 4. The first activity is text analysis (more fully described by Edgington et al [9]), where the underlying linguistic structure is determined and a phonemic transcription of the text is produced. Next come the parallel processes of prosody synthesis and speech sound selection. Prosody synthesis generates the intonation and rhythm of the passage to be synthesised, while speech sound selection involves choosing speech fragments from a recorded speech database. Finally the speech waveform is generated by smoothly joining the selected fragments of recorded speech and imposing the prosody on to it using speech modification algorithms. Speech and prosody generation are discussed in detail by Edgington et al [10].

Stated simply, there are two problems facing text-to-speech system developers — technical constraints and theoretical unknowns. With the recent increase in computing power and the availability of cheap storage, many of the technical constraints have been solved or at least significantly eased. However, there has not been an equivalent increase in our understanding of the production and perception of speech. A number of the latest developments in text-to-speech have tended towards encoding ignorance rather than attempting to model the underlying mechanics. This has led to a divergence in system design, those systems which attempt to understand and model the speech signal (e.g. formant synthesis) and those which attempt to encode significant aspects of the speech signal (e.g. concatenative synthesis). Neither approach has, as yet, been shown to be clearly superior to the other.

By far the most significant advance in the generation of synthetic speech in recent years has been as a result of the impact of low-cost, high-volume storage media, combined with efficient speech modification algorithms. This combination of technology gave a step increase in the perceived naturalness of synthetic speech, with the added bonus that different speakers could be modelled comparatively easily. For the first time speech researchers were able to assess the importance of the prosodic structure of language without being encumbered by the comparatively poor segmental quality which masked significant effects. It is now clear that high-quality natural-sounding speech synthesis will not be achieved through the simple concatenation of segments of sampled speech data. The acceptable segmental quality of the current generation of speech modification algorithms has forced researchers to address significant problems in the specification of synthetic prosody in speech synthesis.

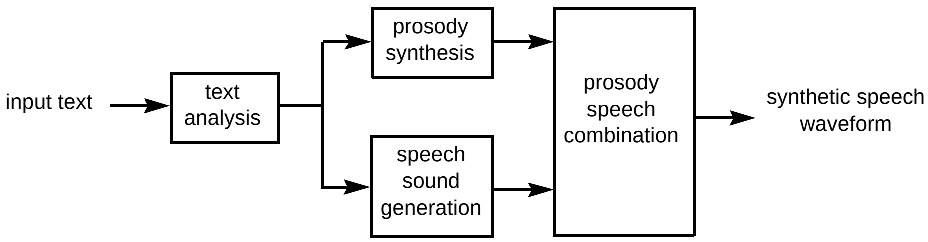

Fig 4 Text-to-speech conversion system.

3.2 Laureate system architecture

The Laureate text-to-speech system has been designed specifically with the aim of allowing a number of different models of spoken language to co-exist within the same computational framework. The basic design premise of Laureate is shown in Fig 5. The different-sized balloons represent different shades and types of linguistic theory, which in isolation are incompatible with each other. The purpose of the Laureate core and component interface is to provide a consistent and sufficient minimum set of standard linguistic representations which are common across the different models and production methods, so allowing different theories to coexist.

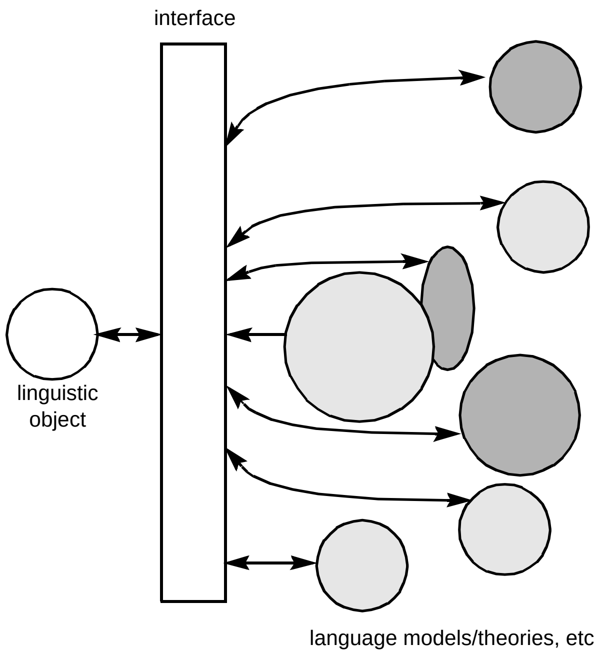

Fig 5 Laureate's basic design premise.

There are a number of potential problems in mixing different theories:

- each theory will require its own internal data structures,
- different theories may have different and even conflicting linguistic definitions,
- different theories will require different amounts of given knowledge.

Figure 6 shows diagrammatically how the Laureate architecture addresses these problems. The Laureate system consists of two main parts, a core component and a number of satellite components. The core represents the backbone of the system. All information within the system passes from one component to another via the linguistic object contained within this core. The linguistic object is a dynamic database of information, which has a simple understanding of the relative hierarchical structure of language.

Satellite components can only access the object via a set of formalised linguistic questions or statements. This means that each satellite component is completely isolated from the rest of the system. It may therefore have its own internal representational linguistic structure, which need not necessarily conform to that of any other component within the system. The formalised linguistic interface ensures that data returned to the linguistic object can be used by other components of the system in a consistent manner. Due to the modular nature of the system, components may be placed at different points along the backbone. Thus, components that require large amounts of linguistic knowledge can be placed after components that add such knowledge to the linguistic object.

The linguistic object expands and contracts as text blocks are processed by the system. A text block represents the minimum amount of text which contains sufficient linguistic information for the system to function correctly. Text blocks are determined within the text normalisation component of the system and may vary in size from single words to paragraphs. As a text block cascades through the system, each satellite component loads information into the linguistic object. All components, with the exception of the realisation component, are only concerned with adding information to the object. Once the realisation component has completed its tasks, all information in the linguistic object will be cleared in preparation for the next text block.

The linguistic object is, in effect, a relational database which has no understanding of the information contained

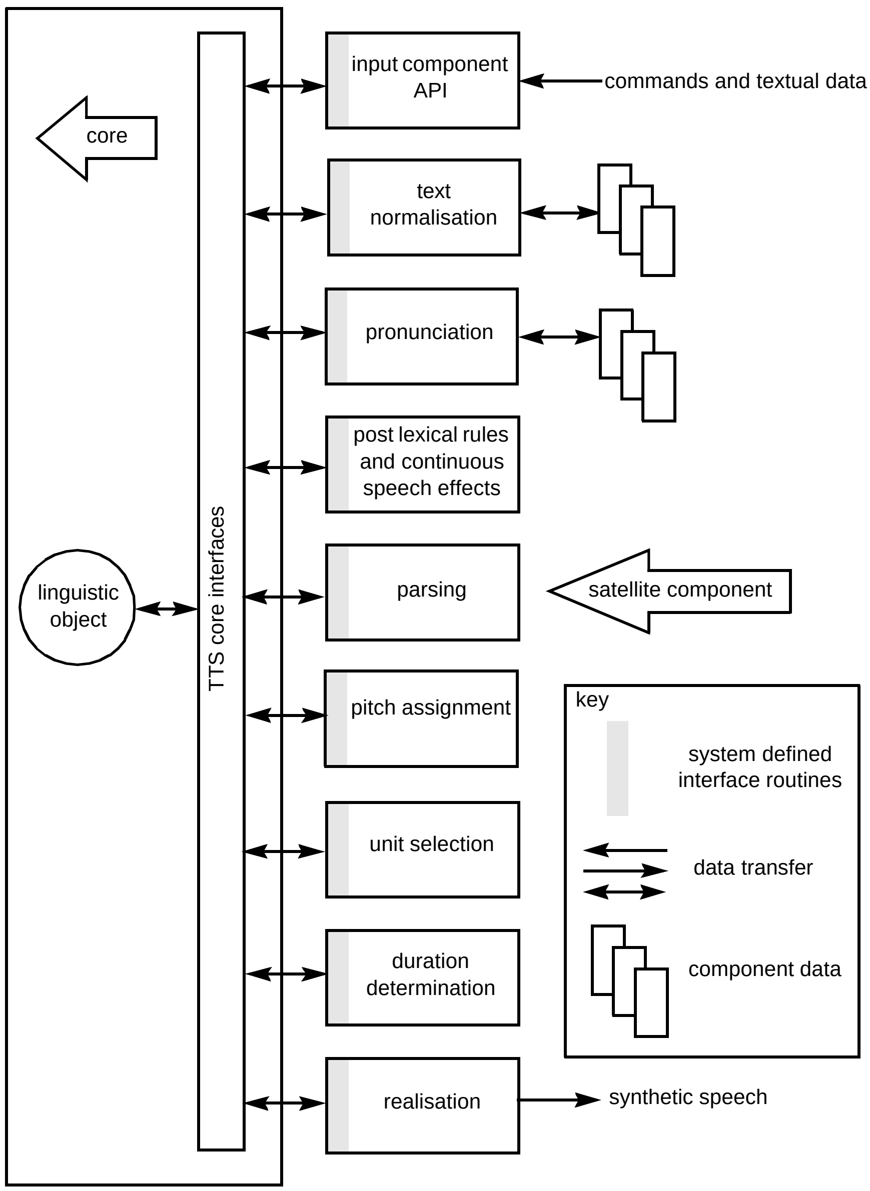

Fig 6 The Laureate architecture.

within each database record, but only allows certain linguistic relationships between records to exist, for example, ‘words contain syllables’. This relationship is reflected in the way word and syllable records relate. As the specific meaning of a piece of information contained in a record may change as satellite components develop, the meaning of record elements is stored externally to the linguistic object so that it is common to all components.

As a practical example of the above, consider the generation of intonation within the Laureate text-to-speech system. There are a number of different theories on how best to specify and generate an intonation contour. Two prominent models are the Fujisaki model and the Silverman [11] model. Basically, the Fujisaki model consists of two components — an accent component and a phrasal component. Values for both of these components must be specified at the outset. A contour is generated by superimposing accent commands of different length and height, on a basic phrasal component which extends over the total length of the phrase. In contrast, the Silverman model views intonation as a realisation of a linear sequence of tonal accents, and so does not contain a phrase level component. A contour is generated by interpolation between specified tones. The two theories clearly make very different assumptions about the processes leading to the generation of intonation, but the Laureate system can accommodate either model as follows:

- two components are commonly used in the generation of intonation within Laureate — the pitch assignment component and the realisation component,

- the pitch assignment component generates the accent information needed to generate a contour,
- the realisation component generates the actual pitch values as dictated by the accent information found in the linguistic object.

Thus, either theory can be accommodated in the Laureate system by simply including the appropriate satellite components. While the meaning of the information contained in the linguistic object has changed, structurally the systems are identical.

Another advantage of the theory-independent structure of Laureate is that it is also a language-independent structure. This minimises the changes required to implement synthesis in alternative languages. Simplistically the structure of Laureate can be viewed as consisting of three parts, as in Fig 7 — the linguistic object (data store), the satellite components (language processing engines) and the language-specific data. The meaning of information stored in the linguistic object is stored as part of the language-specific data. By separating the data from the linguistic object the Laureate architecture provides a structure which can be more easily tailored to different accents and languages. The aim of the linguistic object is to provide a language-independent store, while the aim of the components is to manipulate language-specific data in such a way as to provide new information which can be stored in the linguistic object.

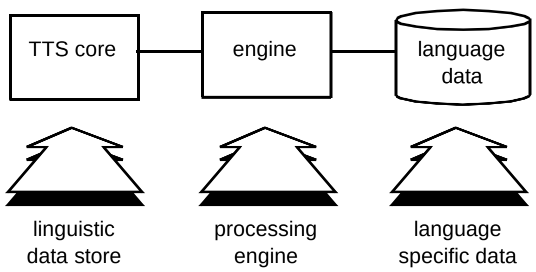

Fig 7 Relationship between linguistic datastore, processing components and language-specific data.

In practice, it is impossible to generate completely language-independent components, but this architecture ensures the maximum reuse of code. For example, if a satellite component needs to be completely replaced this can be done simply without significantly affecting the other components. If extra components are required these may be inserted without modification to existing ones.

Laureate design features

The Laureate system has been designed to meet the challenge of developing a software suite, which can act as an effective research tool and provide the basis for robust and cost-effective applications. The list given below highlights some of the design characteristics provided within the Laureate architecture to achieve this aim.

- Modular design — this, coupled with the concept of a linguistic object, means that each component can be developed in isolation. Consequently the cost of upgrading the system once it has been downstreamed to a platform is greatly reduced as the cost of downstreaming a new or improved component is predominantly the cost of the component testing.
- Execution sequence flexibility — the ability to change the order of execution, insert or remove components without the need for recompilation provides a flexible structure, which will be able to accommodate new ideas as they appear and extend the lifetime of the code.
- Industry standard software — the code has been written in ANSI standard C, which is available on most hardware platforms. Languages other than ‘C’ may be included in the system provided they are capable of linking with ‘C’. For extra flexibility, each satellite component in the system may be configured without recompilation to run as a separately scheduled task.
- API simplifies integration into applications — the Laureate system includes an API (application programmers interface) which eases integration with application software. The applications developer may view the Laureate system as a black-box component.
- Multichannel — the code has been designed to support multichannel operation. Multiple instances of the Laureate system may exist simultaneously within the same or different applications. An application may choose which instance of the Laureate system to use at any one time. The Laureate API code will manage all system resources required to support multiple channels as efficiently as possible.

The linkage of a flexible, portable code structure to a clearly motivated and eclectic theoretical design enables the Laureate code to meet the needs of industry, while acting as an effective research and development tool.

3.3 Tailoring Laureate for specific applications

New voices, accents and languages

The Laureate technology and architecture, permits new voices, accents, and languages, to be incorporated into the system with minimal changes. New voices are introduced by recording the voice of the chosen speaker reading a specially devised script, which has a good coverage of the sounds of the language. This recorded voice data is phonetically annotated before incorporation into Laureate. For regional accents a new voice is required, but depending on the characteristics of the accent, modifications to the pronunciation (lexicons and rules) and pitch assignment systems may also be necessary. A different language also requires changes in areas such as text normalisation and parsing. The core system does not require changes for any accent or language. Some languages and accents will only require different data files. The processing components, which, in general, have been designed to be data-processing engines operating on rules and data, may occasionally require modifications where language-specific features have been unavoidable.

Application specific data

The Laureate text-to-speech system has been designed to enable ease of tailoring to specific applications. To this end its behaviour can be significantly altered simply by changing data files. Wherever possible these are text based, making it easy to change or enhance the content. The Laureate system has the following main types of data file:

- configuration file,
- abbreviation dictionary,
- acronyms dictionary,
- pronunciation dictionary,
- pronunciation rules,
- speech files.

The configuration file determines which of the other data files are used as well as controlling the behaviour of the pronunciation, parser and intonation components. The speech output file format is also selected in the configuration file.

The abbreviation dictionary can be significant to the behaviour of the system in certain types of application, as it controls the way abbreviations are expanded. Abbreviations can pose problems for a text-to-speech system as they rarely have a unique expansion. The common examples of ‘st.’ and ‘dr.’ are good examples as they both commonly have two expansions: Saint and street for ‘st.’ and doctor and drive for ‘dr.’ Intelligence is built in to the text normalisation component of Laureate to resolve these common ambiguities [12]. However, in many specialist areas abbreviations are reused with their own local meanings in a way that is impossible to disambiguate. In these situations a specialist abbreviations dictionary must be generated just covering the area of the application.

A dictionary is provided for acronyms that are pronounced as a complete word such as ‘DEC’, and ‘ASCII’.

Pronunciation dictionaries can also be tailored to the application. An example of when this is useful is in an interactive voice response (IVR) system, which confirms a name by reading it back using synthesis. In this situation the only option is to have large name dictionaries to get good synthesis performance, as names in the UK are notoriously difficult to pronounce by rules. However the Laureate system uses a specialist technique for name pronunciation called ‘pronunciation by analogy’ which can perform significantly better than normal English pronunciation rules.

4. Laureate performance

Measuring the performance of text-to-speech systems is not a straightforward matter. An obvious measurement is intelligibility, which did provide a useful measure for the early systems. The quality of today’s text-to-speech systems are still not as good as real speech, but most systems are intelligible most of the time. Consequently, a measurement is needed that can extract the differences between systems that have a performance between basic intelligibility and high-quality recorded speech. The techniques used by BTL to measure the performance of speech synthesisers are based on methods that have been devised over many years for the measurement of the performance of telephone system components. These tried and tested techniques have been chosen for their reliability and independence from the ability of the subject rather than the performance of the TTS system under consideration.

One of the factors often overlooked in this type of testing is the sound level. This has a significant effect on the ‘listening effort’ and therefore measurements must be done over a range of levels for meaningful results to be obtained. The subjects are asked to rate the speech in terms of the effort required to understand the meaning of sentences. They are given a five-point scale, where the points have the following descriptions:

1 complete relaxation possible — no effort required,

2 attention necessary — no appreciable effort required,

3 moderate effort required,

4 considerable effort required,

5 no meaning understood with any feasible effort.

Measurements obtained in this way are reliable and consistent, when used in combination with a reference system as shown by Johnston [13]. The performance of Laureate and many leading commercial speech synthesis systems have been measured using these techniques. At normal listening levels the performance of Laureate is better than any other system tested and this is shown in the graphs of two different performance tests against four other commercial synthesis systems (see Figs 8 and 9).

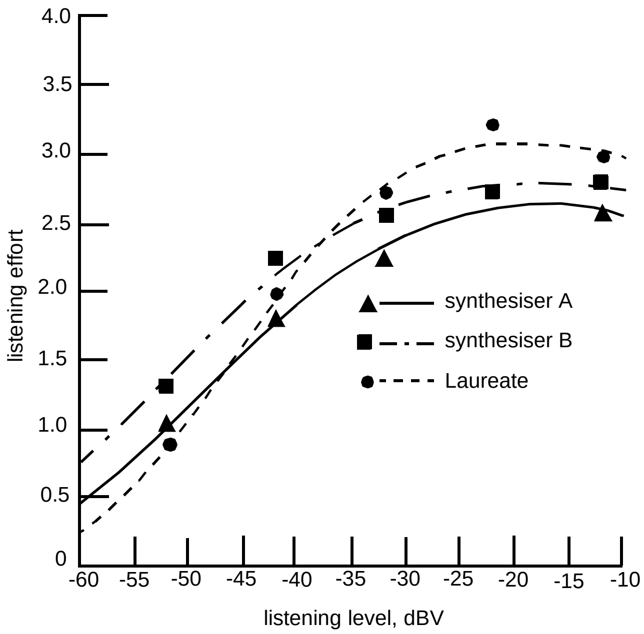

Fig 8 Sample synthesiser performance test results.

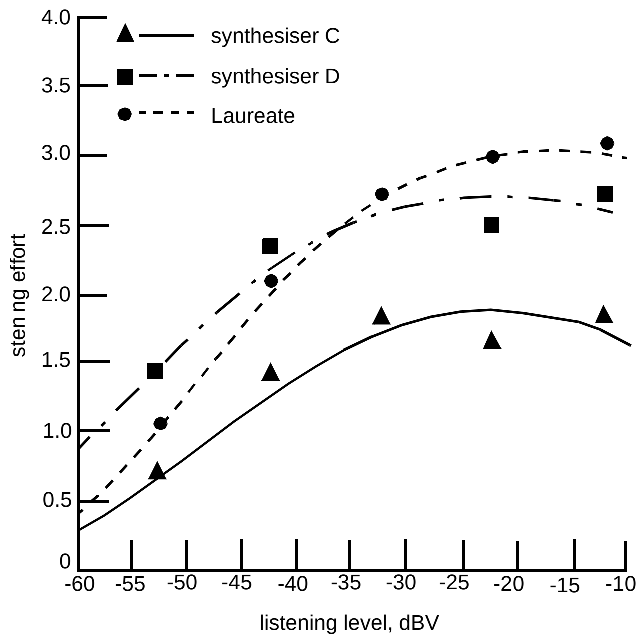

Fig 9 Sample synthesiser performance test results.

5. Conclusions

Text-to-speech synthesis is generating increasing interest due to its ability to transform information from one medium to another. This is of growing importance in the move towards a truly multimedia world, where people will want information presented in the medium of their choice. The possibilities for applications delivering computer-based textual information over telephone networks are almost without limit. This makes text-to-speech synthesis of great importance in an information age, where the major form of communication used by customers is still speech using the ubiquitous telephone. This paper has given examples of some applications where text-to-speech synthesis can currently be used to greatest benefit. The unique design philosophy and structure of the Laureate text-to-speech system, which can support both downstreaming and continuing research, has been presented, as well as the ways in which the Laureate synthesis system can be tailored to applications.

The Laureate system has been easily integrated into a number of different prototype systems which include telephone access to e-mail, weather forecast, road traffic congestion and medical information lines, and WWW pages. It has also been ported to a number of Unix ™ workstations and personal computers as well as to the BT speech applications platform [14] where other speech processing algorithms can be called on by applications programmers building speech-based services.

Two companion papers describe the modern techniques used to perform the conversion of text into speech. The first deals with text analysis [9] and the second, prosody and speech generation [10].

Finally, with the accelerating advances in synthesis technology, the predictions of Arthur C Clarke may well be fulfilled, and a speaking machine may yet sound as convincing as HAL by the year 2001.

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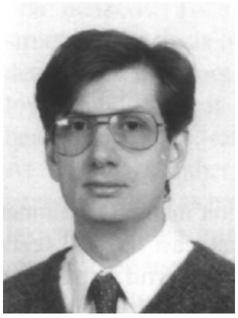
Julian Page graduated from Warwick University in 1972 with a BSc honours degree in engineering science. He then joined BT and worked in a group designing submarine coaxial cables several of which were installed across the Atlantic.

In 1979 he spent a short period in a group responsible for the design of digital PABXs. In 1980, he moved again, on promotion, to a group responsible for the design of high-speed voice band modems where he was involved in the design of several successful modem products, as well as having responsibility for the BT datapump project, a key component in the BT Fifth Generation range of modems.

He is currently leading a team undertaking R&D on speech synthesis and analysis techniques, the main output of which has been the Laureate text-to-speech synthesiser.



Andrew Breen holds a BSc in Physics with Computing Physics from University College Swansea, an MSc (Eng) in Electrical Engineering from Liverpool University and a PhD in Speech Science from University College London. After completing his first degree he was employed in the department of Electronics and Electrical Engineering at Liverpool University where he worked for three years in the area of speech coding and digital filter design. After leaving Liverpool University he was employed in the department of Phonetics and Linguistics at UCL where he worked for two years on continuous speech recognition before moving into the area of text-to-speech synthesis. He is project leader of the language component of the Laureate TTS system and project manager of the Synthetic Persona project. His current research interests include speech synthesis, speech prosody and the representation of linguistic and para-linguistic information in a synethetic persona. He is a member of the IEE.